Career Guide

Amazon Data Analyst Interview Questions 2026: SQL, Cases & Solutions

Comprehensive 2026 guide to Amazon data analyst and analytics interviews. Round breakdowns, live SQL problem scenarios with code solutions, and compensation bands.

Anuj Saini Sep 3, 2026 14 min read

Amazon Data Analyst Interview Guide (Quick Summary)

KEY TAKEAWAY

Amazon Data Analyst and BIE interviews in 2026 emphasize metric ownership, data modeling, and Leadership Principles. The technical loop tests complex SQL (window functions, CTEs, self-joins), business metrics intuition (Customer LTV, Prime churn), and root-cause diagnostic capabilities.

Cracking a Data Analyst or Business Intelligence role at **Amazon** requires more than memorizing basic SQL syntax. As one of the premier employers in the

Cloud, E-Commerce & Retail space, Amazon's interview loops are designed to test your ability to translate ambiguous business problems into rigorous, optimized queries and actionable product insights.

In this comprehensive 2026 preparation guide, we break down the **4-stage interview process**, explore **core SQL problem patterns with working PostgreSQL solutions**, analyze **key product metrics**, and review **verified compensation benchmarks**.

The Amazon Data Analyst Interview Process (4 Rounds)

Amazon structures its analytics hiring loop into four distinct stages:

Round 1: Online Technical Assessment (75 Mins)

HackerRank assessment with 3 SQL problems testing window functions (RANK, LAG/LEAD), multi-table joins, and 1 data modeling scenario.

Round 2: Technical SQL & Data Modeling (60 Mins)

Live coding with a BIE on dimensional schemas (fact vs dimension tables, star schema), handling high-scale cardinality, and query optimization.

Round 3: Business Analytics & Metric Deep Dive (60 Mins)

Focuses on metric definition: defining North Star metrics for Prime Video or Amazon Fresh, designing A/B experiments, and diagnosing funnel dips.

Round 4 & 5: Leadership Principles & Bar Raiser (2x 45 Mins)

Intense behavioral evaluation using the STAR framework centered on Customer Obsession, Ownership, Dive Deep, and Bias for Action.

Top Amazon SQL Interview Problems & Solutions

Here are realistic SQL problems reflecting the exact domain shapes tested in Amazon technical rounds:

Problem 1: 80/20 Pareto Revenue Contribution Analysis

Business Scenario:
Amazon marketplace analytics needs to identify the top tier of sellers that generate the cumulative first 80% of total marketplace gross merchandise value (GMV).

Table Schema:

SQL Copy code

```
sellers(seller_id, total_gmv)
```

Optimal SQL Solution (PostgreSQL):

SQL Copy code

```
WITH running_totals AS (
    SELECT
        seller_id,
        total_gmv,
```

```sql
        SUM(total_gmv) OVER (ORDER BY total_gmv DESC) AS running_gmv,
        SUM(total_gmv) OVER () AS total_platform_gmv
    FROM sellers
)
SELECT
    seller_id,
    total_gmv,
    ROUND((running_gmv / total_platform_gmv) * 100, 2) AS cumulative_pct
FROM running_totals
WHERE (running_gmv - total_gmv) / total_platform_gmv < 0.80
ORDER BY total_gmv DESC;
```

Step-by-Step Logic Breakdown:

1. `SUM(total_gmv) OVER ()` calculates the grand total in one pass.
2. `SUM() OVER (ORDER BY total_gmv DESC)` computes the descending running total.
3. The `WHERE` clause filters the threshold for the 80% boundary.

> ⚠ **Common Candidate Pitfall**
>
> Using correlated subqueries instead of window functions, causing quadratic execution slowdown on large datasets.

> *[!TIP]* ***Test your solution in Topfolio's SQL sandbox:*** *Write and run real SQL queries against this exact schema with instant PostgreSQL grading.*
>
> 👉 ***Solve this question live in the Interactive Sandbox →***

Problem 2: Customer Spend Quartiles with NTILE

Business Scenario:

Divide Amazon Prime customers into 4 equal quartiles based on their annual spend to tailor promotional discount tiers.

Table Schema:

SQL Copy code

```
customer_orders(customer_id, annual_spend)
```

Optimal SQL Solution (PostgreSQL):

SQL Copy code

```sql
SELECT
    customer_id,
    annual_spend,
    NTILE(4) OVER (ORDER BY annual_spend DESC) AS spend_quartile,
    CASE NTILE(4) OVER (ORDER BY annual_spend DESC)
        WHEN 1 THEN 'Platinum (Top 25%)'
        WHEN 2 THEN 'Gold (25-50%)'
        WHEN 3 THEN 'Silver (50-75%)'
        WHEN 4 THEN 'Bronze (Bottom 25%)'
    END AS tier_label
FROM customer_orders;
```

Step-by-Step Logic Breakdown:

`NTILE(4)` partitions the ordered dataset into four equal frequency buckets, assigning quartile ranks 1 through 4.

⚠ **Common Candidate Pitfall**

Assuming `NTILE` splits by spend amount rather than row count. NTILE guarantees equal row distribution across buckets.

> *[!TIP]* ***Test your solution in Topfolio's SQL sandbox:*** *Write and run real SQL queries against this exact schema with instant PostgreSQL grading.*
> 👉 ***Solve this question live in the Interactive Sandbox →***

Key Product Metrics & Case Studies at Amazon

In the product sense round, interviewers will ask you to define metrics and diagnose anomalies. Be prepared for questions such as:

1. **North Star Metric Definition**: What is the primary North Star metric for Amazon's core business vertical, and what are 2 counter-metrics to ensure quality is not sacrificed for growth?
2. **Funnel Drop-Off Diagnosis**: *"Conversion dropped by 8% week-over-week in Bengaluru. How would you structure your diagnostic investigation?"*
3. **A/B Experiment Design**: How would you design an experiment to test a new recommendation algorithm without cannibalizing organic merchant search?

2026 Amazon Data Analyst Salary Bands (India)

Based on verified market submissions, here is how compensation is structured at Amazon:

LEVEL / EXPERIENCE	FIXED BASE CASH	ANNUAL PERFORMANCE BONUS	RSUS / ESOPS (PER YEAR)	TOTAL ANNUAL CTC
Entry Level (0–2 Years)	₹10.0 – ₹15.0 LPA	₹1.0 – ₹2.5 LPA	₹1.0 – ₹3.0 LPA	**₹16–24 LPA**
Mid Level (2–5 Years)	₹18.0 – ₹28.0 LPA	₹2.5 – ₹4.5 LPA	₹3.0 – ₹7.0 LPA	**₹28–45 LPA**
Senior / Lead (5+ Years)	₹32.0 – ₹50.0 LPA	₹5.0 – ₹10.0 LPA	₹10.0 – ₹25.0+ LPA	**₹55–90+ LPA**

*For a detailed breakdown of how Amazon compares to IT Services, GCCs, and Tier-1 tech firms, see our comprehensive **Data Analyst Salary Guide 2026** and the **India Data Roles Salary Progression Guide**.*

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Frequently Asked Questions

How many interview rounds are there for a Data Analyst at Amazon?

Amazon data analyst interviews typically consist of 4 rounds: Online SQL & Aptitude Screening, Live Machine Coding / Data Wrangling, Product Sense & Case Diagnostics, and a Hiring Manager / Cultural Fit round.

What SQL topics are most heavily tested at Amazon?

The technical bar heavily focuses on Business Intelligence (BIE), metric ownership, 80/20 Pareto analysis, customer lifetime value tiers, and Leadership Principles.

What is the average Data Analyst salary at Amazon in India in 2026?

At Amazon, entry-level Data Analysts (0–2 years) earn ₹16–24 LPA, mid-level analysts (2–5 years) earn ₹28–45 LPA, and senior analysts (5+ years) earn ₹55–90+ LPA in total compensation.

How can I practice Amazon-style SQL interview questions for free?

Topfolio provides free, interactive in-browser SQL practice questions tailored to Amazon's problem shapes with instant PostgreSQL database grading and step-by-step solutions.



WRITTEN BY

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Founder at Topfolio with 6+ years in data & analytics across JPMC, Ultrahuman, and high-growth startups. Sat on hiring panels, reviewed 500+ resumes, and writes practical SQL & data guides.

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